

P2P Market Monitor

Final QA Report

QA iteration: 19 September 2026

Scope: current implemented build - live V1 monitoring, web UI, Telegram path, V2 historical collector and PostgreSQL persistence.

Iteration status	Retest	Targeted regression	Major open defect
COMPLETED	PASS	PASS	BUG-001 OPEN

Bottom line

The main implemented user flow remained functional after the selected fixes. Input-validation and active-monitor UX issues were fixed and verified. One rare V2 long-cycle defect remains open for additional research.

This report preserves the full QA timeline: pre-fix failures, fixes, retest, targeted regression, endurance evidence, open issues, and deferred coverage.

1. Executive summary

The QA iteration used smoke, functional, negative, exploratory, endurance, research, retest, and targeted regression activities. The selected reproducible validation/UI issues were fixed and verified. The iteration is closed for the current selected scope, while the rare V2 long-cycle issue remains open.

Testing scope

- Web UI startup and monitoring flow.
- User input handling and validation.
- Bybit P2P request/response handling and live monitoring behavior.
- Merchant/compatibility filtering and best-order selection.
- Cached USD/RUB reference usage and edge calculation.
- Telegram notification path.
- V2 pagination, historical snapshot collection and PostgreSQL persistence.
- Retry/timeout behavior, long-running stability, retest and targeted regression.

Out of scope

- Trade execution and real-money transaction processing.
- Production deployment and multi-user production load testing.
- Full security audit.
- Future V2 support scoring and subscription/billing features.

2. Test coverage and status

Test area	Status	Result
Smoke	PASS	Critical demo path exercised
Functional	PASS / PARTIAL	Core path worked; some standalone checks skipped
Negative	MIXED	Defects, UX findings and requirement gaps discovered
Exploratory	COMPLETED	Two active-monitor findings discovered and later fixed
Endurance - V1	PASS	12h 34m 56.6s; retries and stale-response path exercised
Endurance - V2	ISSUE FOUND	BUG-001 discovered
Research	IN PROGRESS	Two network scenarios did not reproduce BUG-001
Retest	PASS	Selected fixes verified

Test area	Status	Result
Targeted regression	PASS	Neighboring changed-area behavior remained functional
Full regression suite	NOT CLAIMED	Not every planned regression item was executed

Smoke checks recorded as passed

- Reference path
- V1 runner
- Web UI
- Telegram alert path
- V2 runner
- PostgreSQL snapshot save

Functional precision

The report does not convert every planned checklist item into PASS. Pagination, snapshot persistence, order-count consistency, reference persistence and stored order review were exercised; standalone timestamp-consistency validation was skipped.

3. Negative and exploratory findings

Amount input

- Negative amount: PASS - rejected.
- Fractional amount: PASS - rejected.
- Empty amount: PASS - intentionally supported as 'any amount'.
- Zero amount: FAIL before fix; later fixed and retested PASS.
- Extremely large amount: requirement gap - no MAX_AMOUNT defined.

Target profit

- Fractional values: PASS.
- Letters: PASS - blocked.
- Negative zero / zero formatting: FAIL before fix; later normalized to 0.
- Explicit leading plus: accepted before fix; later blocked under the chosen product rule.
- MIN/MAX accepted range: requirement gap.

Exploratory session

- Optional fields remained editable during active monitoring - UX/state-consistency issue.
- START button did not reflect active monitor state - UX issue.
- Timer worked correctly.
- Matching orders displayed correctly.

- Page refresh preserved active monitor state.

The two selected active-monitor UX findings were fixed and verified in the same iteration.

Network loss during V1

Core recovery behavior passed: the process stayed alive, retry/error paths activated, no false Telegram alert was observed, and valid results resumed after reconnection. A remaining UX improvement is to expose temporary connectivity/retry status in the UI.

Missing reference

This remains open. The user can reach an active-looking monitoring state without clear UI feedback that the reference value is unavailable.

4. Endurance evidence

V1 long-run result

Metric	Value
Observed uptime	12h 34m 56.6s
Requests launched	9031
Requests completed	9030
Requests succeeded	9030
Requests failed	0
Results processed	9025
Stale responses dropped	1
Telegram notifications sent	100
Bybit retry events in raw log	2

The long-running V1 monitor remained operational through the observed retry events. One delayed response was discarded as stale after a fresher request had already completed.

V2 normal baseline

Metric	Baseline cycles 1-27
Pages per snapshot	4
Collection time	35.191s - 40.957s
Mean collection time	37.830s
Median collection time	37.415s
Orders per snapshot	295 - 329
Mean orders	314.5

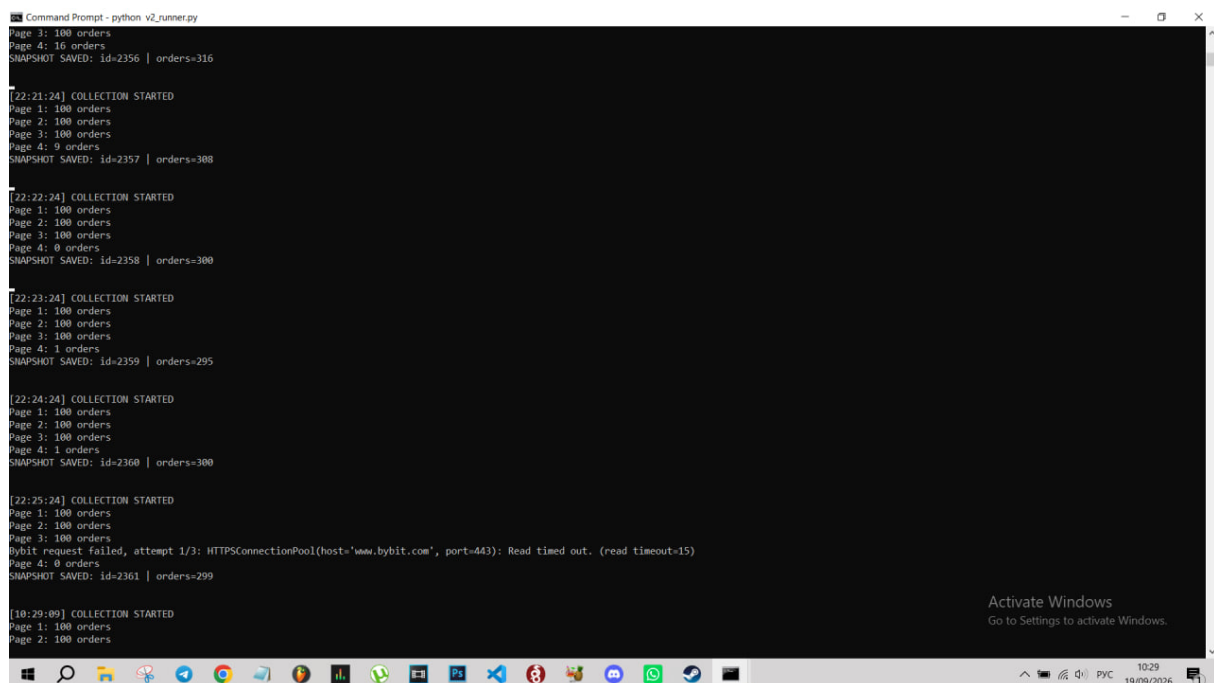
5. BUG-001 - V2 abnormal long cycle

Status: OPEN / RESEARCH REQUIRED

- Cycle 28 recorded a Bybit retry on page 4.
- Snapshot 2361 was still saved with 299 orders.
- Collection time: 43424.645s.
- Total cycle time: 43424.824s.
- CYCLE_OVERRUN: approximately 12 hours.
- Cycle 29 recovered automatically and completed in 29.762s.

Impact

- No fresh historical snapshot is produced while the cycle is stuck.
- One saved snapshot can span a very large real-time interval.
- Market orders and reference context may no longer describe one coherent market state.
- Downstream analysis may treat the abnormal snapshot as ordinary data unless filtered.



```
Command Prompt - python v2_runner.py
Page 3: 100 orders
Page 4: 16 orders
SNAPSHOT SAVED: id=2356 | orders=316

[22:21:24] COLLECTION STARTED
Page 1: 100 orders
Page 2: 100 orders
Page 3: 100 orders
Page 4: 9 orders
SNAPSHOT SAVED: id=2357 | orders=308

[22:22:24] COLLECTION STARTED
Page 1: 100 orders
Page 2: 100 orders
Page 3: 100 orders
Page 4: 0 orders
SNAPSHOT SAVED: id=2358 | orders=300

[22:23:24] COLLECTION STARTED
Page 1: 100 orders
Page 2: 100 orders
Page 3: 100 orders
Page 4: 1 orders
SNAPSHOT SAVED: id=2359 | orders=295

[22:24:24] COLLECTION STARTED
Page 1: 100 orders
Page 2: 100 orders
Page 3: 100 orders
Page 4: 1 orders
SNAPSHOT SAVED: id=2360 | orders=300

[22:25:24] COLLECTION STARTED
Page 1: 100 orders
Page 2: 100 orders
Page 3: 100 orders
Bybit request failed, attempt 1/3: HTTPSConnectionPool(host='www.bybit.com', port=443): Read timed out. (read timeout=15)
Page 4: 0 orders
SNAPSHOT SAVED: id=2361 | orders=299

[10:29:09] COLLECTION STARTED
Page 1: 100 orders
Page 2: 100 orders
```

Terminal evidence: normal minute-scale cycles, a page-4 timeout/retry, the delayed snapshot, and the next collection after the long gap.

6. Research testing after BUG-001

Experiment 1 - network loss through retry exhaustion

Result: NOT REPRODUCED. The collector failed/retried and the next scheduled cycle started; the 12-hour condition did not recur.

Experiment 2 - connectivity restored before request timeout

Result: NOT REPRODUCED. The active request completed without an application-level retry/error, and collection continued normally.

Research conclusion

A simple disconnect/reconnect is not sufficient to reproduce the original condition. The next useful step is observability rather than more random network toggling.

Deferred diagnostic instrumentation

- REQUEST_START / REQUEST_END per page.
- NORMALIZE_START / NORMALIZE_END.
- REFERENCE_START / REFERENCE_END.
- DB_SAVE_START / DB_SAVE_END.
- Optional CYCLE and RETRY start/end markers.

7. Fix verification

Retest: PASS

- amount=0 rejected; empty amount remains supported.
- Active optional fields are locked.
- Active control reflects monitoring state and supports Stop.
- UI returns to ready state after stopping.
- Zero-equivalent target-profit values normalize to 0.
- Explicit leading plus is blocked.
- Negative and fractional target-profit values remain supported.

Targeted regression: PASS

- Valid amount values still work.
- Negative/fractional amount validation remains intact.
- Monitoring can start, stop, and start again.
- Timer and refresh behavior remain correct.
- Second-tab monitor state remains consistent.
- Valid positive, negative, and fractional target-profit values still work.
- Critical V1 / Bybit / reference / result / Telegram path remains functional.

Claim boundary

This was targeted regression around the changed areas. A full project-wide regression suite was not executed and is not claimed.

8. Open items and deferred coverage

Item	Status
BUG-001 V2 long cycle	OPEN / RESEARCH REQUIRED
Missing reference status in UI	OPEN
Temporary network/retry status visibility	OPEN UX improvement

Item	Status
24h no-result auto-stop/message/reset	PLANNED PRODUCT LOGIC
Maximum accepted amount	REQUIREMENT GAP
Target-profit MIN/MAX range	REQUIREMENT GAP
Telegram failure injection	NOT EXECUTED
Database failure/rollback test	NOT EXECUTED
Malformed Bybit payload fail-closed test	NOT EXECUTED

9. Final conclusion

Current QA iteration: COMPLETED

- Retest: PASS.
- Targeted regression: PASS.
- Main implemented user flow: functional after selected fixes.
- BUG-001: remains open and requires additional diagnostic research.
- Pre-fix FAIL evidence is preserved rather than rewritten, keeping the QA process traceable.

This package is suitable as a portfolio QA case because it demonstrates the complete loop: plan -> execute -> find defects/gaps -> investigate -> fix selected issues -> retest -> targeted regression -> preserve open-risk evidence.